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基于选择性对抗运动先验的强化学习用于人形机器人多步态学习

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摘要——在统一的强化学习框架中，让人形机器人学习多样化的运动技能仍然具有挑战性，因为不同步态对稳定性和动态表现力的要求相互冲突。我们提出了一种多步态学习方法，通过一致的策略结构、动作空间和奖励函数设定，使人形机器人能够掌握五种不同步态：行走、正步走、跑步、爬楼梯和跳跃。关键贡献在于一种选择性对抗运动先验（***AMP***）策略：***AMP***被应用于周期性且稳定性关键的步态（行走、正步走、爬楼梯），以加速收敛并抑制不规则行为；而对于高动态步态（跑步、跳跃），则刻意不使用***AMP***，因为其正则化会过度约束运动。策略通过***PPO***算法在仿真环境中结合域随机化进行训练，并通过零样本仿真到现实迁移部署到一台***12***自由度人形机器人上。定量比较表明，选择性***AMP***策略在所有五种步态上均优于统一应用***AMP***的策略，在不牺牲动态步态所需敏捷性的前提下，实现了更快的收敛、更低的跟踪误差和更高的稳定性相关步态成功率。

I. 引言

类人机器人有望在专为人类设计的环境中运行，但实现这一潜力需要具备自适应地执行多种运动行为的能力 [1]。实际部署不仅需要用于高效通行的行走，也需要用于快速移动的奔跑、用于跨越间隙或越障的跳跃、用于多层环境导航的爬楼梯，以及面向规整队形或仪式场景的正步走等专用步态。每种步态对控制系统都提出了根本不同的要求：行走强调能效与稳定平衡，奔跑要求快速产生力并管理腾空阶段，跳跃要求在极短时间内产生最大垂直冲量，爬楼梯需要精确的落脚点和离地间隙控制，正步走则需要持续的腿部伸展和严格的姿态约束。核心挑战并非孤立地学会每种步态，而是在共享策略架构、观测空间和奖励结构的统一框架中学习它们。

强化学习已成为一种在没有显式解析模型的情况下学习复杂运动技能的极具吸引力的方法。通过与物理模拟器进行大规模交互，强化学习智能体能够发现难以手工设计的控制策略，尤其是对于仿人机器人 [2]这类欠驱动、高自由度系统。相较于传统的基于模型的控制方法，强化学习的核心优势在于能够在统一的优化过程中处理非线性动力学、接触切换和执行器限制。近期研究已展现出令人印象深刻的单步态能力，包括在平坦和不平整地面上的稳健行走、带有离地间隙的楼梯通行以及基本的扰动恢复 [3], [4]。然而，这些研究通常为每个任务单独训练一个策略，且每个策略都配有各自设计的奖励和超参数配置。将强化学习扩展到单一框架下的多步态带来了新的难题：策略必须协调不同行为之间相互冲突的动力学；奖励函数必须在运动特性差异巨大的多种步态下仍然保持有效；观测空间必须为所有行为编码足够的信息，同时不变得冗余，也不会对任何单个步态产生误导 [5]。

对抗式运动先验（AMP）解决了基于强化学习的运动控制中的一个关键难题——生成自然、稳定且能良好迁移到硬件的运动 [6]。通过训练判别器来区分智能体生成的运动片段与参考运动片段，AMP 将策略正则化到已展示的类人行为分布上。这已被证明可以减少关节抖动、改善足部落点一致性，并产生更平滑、更符合真实执行器动态的轨迹 [7], [8]。然而，现有采用 AMP 的工作都是统一应用它：一旦引入 AMP 模块，它在整个训练过程中对所有行为始终保持激活。这就引出一个重要但尚未充分探索的问题：AMP 是否总是有益的，或者对某些步态类型是否可能适得其反？

我们的观察是，AMP 的有效性在很大程度上取决于目标步态的性质。AMP 擅长强化周期性的、重复性的运动模式，且关节幅度适中——这正是行走、正步走和爬楼梯所具有的特征。对于这些步态，参考运动分布与期望的策略输出高度匹配，对抗奖励提供了一种稳定的归纳偏置，能够加速收敛并抑制异常行为。相反，对于需要大幅度关节运动、快速加速和爆发性动态的步态——如跑步和跳跃——可用的人体运动参考有限，AMP 判别器的正则化可能成为一种不利的约束。在这些情况下，

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策略需要超越参考流形进行探索，以发现能充分利用机器人执行器能力的策略，而 AMP 惩罚可能会阻止智能体达到所需的运动极限。

这种不匹配促使我们采用一种选择性方法：与其普遍应用AMP或完全弃用，我们建议在它有助益的地方使用它，在它起阻碍作用的地方停用它。这一策略要求理解哪些步态特征符合AMP假设，哪些偏离该假设，并为每种运动模式做出明智的决策。

本文的贡献如下：

1) 一个统一的强化学习（RL）框架，用于学习人形机器人的五种实用步态——行走、正步走、跑步、跳跃和爬楼梯——采用一致的策略架构和基于正弦关节参考跟踪的奖励函数。所有步态共享相同的观测空间、网络结构和奖励公式，仅在参考轨迹参数和奖励权重上有所不同。

2) 一种选择性的对抗运动先验（AMP）策略，将AMP应用于周期性、对稳定性关键的步态，而不用于高度动态、敏捷的步态。我们提供的实证分析表明，AMP提高了行走类步态的收敛速度和跟踪精度，但限制了跑步和跳跃的运动幅度和表现力。

3) 在真实机器人上对全部五种步态进行完整验证，并进行定量比较，包括跟踪误差、收敛速度、成功/跌倒率以及 AMP 与非 AMP 的性能。我们将每项学习到的策略都部署到物理人形机器人平台上，为多步态学习提供了一个可复现的真实机器人基线。

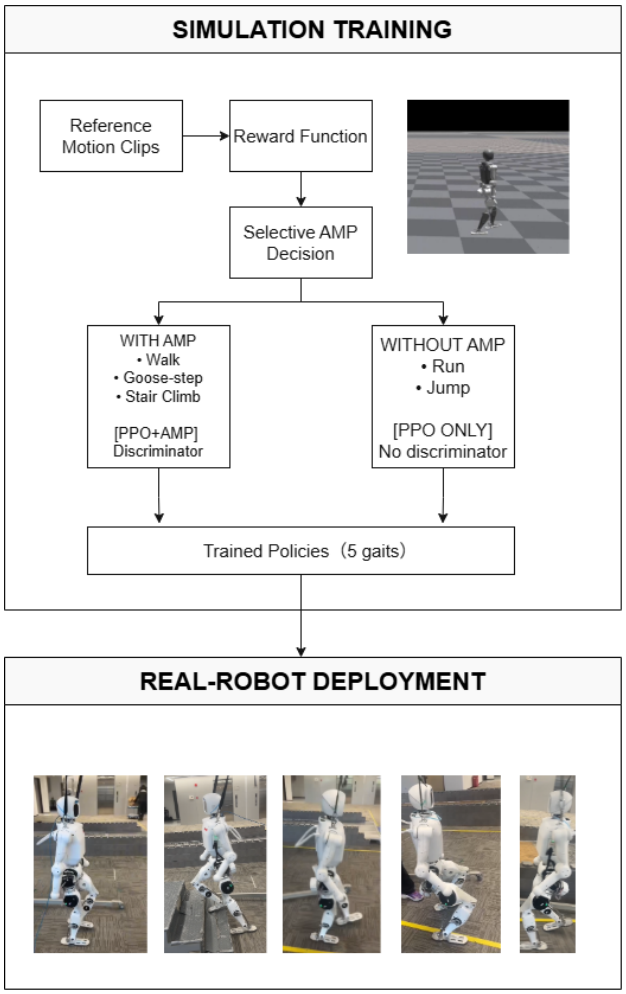


图1. 多步态学习流程概览。根据步态类型，策略在仿真中采用或不采用AMP进行训练，然后部署到物理机器人上。

II. 方法

A. 概述

我们提出了一个统一的强化学习框架，用于让一台 12 自由度人形机器人学习五种不同的运动步态：行走、正步走、跑步、跳跃和爬楼梯。该框架由三个主要部分组成。首先，使用统一的问题设定和一致的观测空间、动作空间及奖励结构来训练所有五种步态，每种步态仅通过其正弦参考轨迹参数和奖励权重配置进行区分。其次，采用选择性 AMP 策略：对于周期性、对稳定性要求高的步态（行走、正步走、爬楼梯）启用 AMP，而对于高动态步态（跑步、跳跃）禁用 AMP。第三，在仿真中通过大规模领域随机化训练得到的策略，无需微调即可直接部署到物理人形机器人上。图 1 展示了整体流程。

B. 强化学习表述

我们将每个步态学习任务形式化为一个由元组 $(\mathcal{S}, \mathcal{A}, r, \gamma, \rho_0)$ 定义的马尔可夫决策过程（MDP）。所有五种步态共享相同的 MDP 结构，不同之处仅在于参考轨迹生成、奖励权重参数以及AMP配置。

1) 状态空间：策略在每个时间步 t 接收的观测空间由本体感觉状态、参考信号和指令输入组成。每帧观测向量 $o_t \in \mathbb{R}^{50}$ 由以下分量拼接而成：

- 基座线速度 $v_b \in \mathbb{R}^3$：在机器人机体坐标系中表示，缩放系数为2.0。
- 速度指令 $c \in \mathbb{R}^3$：目标线速度在 x和 y 方向上，以及绕 z 轴的角速度。
- 相位编码 $\phi \in \mathbb{R}^2$：当前步态相位的正弦编码$[\sin(2\pi\phi_t), \cos(2\pi\phi_t)]$ 。
- 基座角速度 $\omega_b \in \mathbb{R}^3$：在机体坐标系中表示，按0.25缩放。
- 投影重力向量 $g_p \in \mathbb{R}^3$：世界坐标系下的重力向量 $[0, 0, -1]^\top$ 经基座姿态旋转得到。
- 关节位置误差 $q_e \in \mathbb{R}^{12}$：当前关节位置与默认站立配置之间的差异。

- 关节速度 $\dot{q} \in \mathbb{R}^{12}$：按 ***0.05*** 缩放。
- 上一动作 $a_{t-1} \in \mathbb{R}^{12}$：在上一时间步执行的动作。

相位变量（ $\phi_t \in [0,1)$ 按 $\phi_{t+1} = (\phi_t + \Delta t / T_{\text{cycle}}) \bmod 1$ 推进，其中 T_{cycle} 是步态特定的周期时间。

参考关节轨迹通过相位变量进行编码。对于周期性步态（行走、正步走、跑步、爬楼梯），参考轨迹为正弦形式：

$$s = \sin(2\pi\phi_t) \tag{1}$$

$$q_{\text{ref}}^{\text{L,hip}} = -s \cdot \mathbb{1}_{s<0} \tag{2}$$

$$q_{\text{ref}}^{\text{L,knee}} = 2s \cdot \sigma \cdot \mathbb{1}_{s<0} \tag{3}$$

其中 σ 是目标关节位置缩放比例（行走和跑步为 0.26，正步走为 0.28），当 $s > 0$ 时，右腿与左腿镜像对称且符号相反。每种步态通过其 (T_{cycle}，支撑相比例) 参数对调节该模式：行走 (0.8 s, 0.6)、正步走 (0.7 s, 0.45)、跑步 (0.4 s, 0.35)、爬楼梯 (0.7 s, 0.6)。

对于跳跃步态，则使用分段式 4.0 秒周期轨迹，包含五个阶段：下蹲（$\phi < 0.30$，$\sin^2$ 膝关节屈曲）、起跳（0.30–0.42，快速伸展）、腾空（0.42–0.48，保持伸展）、落地（0.48–0.75，冲击吸收与恢复）和站立（$\phi \geq 0.75$，默认姿态）。课程学习会在 2000 至 10000 次 PPO 迭代中逐步加大下蹲深度。

为了捕捉时间动态，策略接收包含 $N = 21$ 个连续观测帧的历史序列，并将其堆叠为维度为 $50 \times 21 = 1050$ 的单一输入。这样就可以在不进行显式导数计算的情况下，隐式估计速度和加速度。

评论网络接收一个扩展观测 s_t，其中额外包含仅仿真可用的特权信息：机体线速度与角速度、投影重力、指令、关节位置误差、关节速度、动作、相位编码、摩擦系数、足部接触掩码、支撑相掩码、关节跟踪误差、推力、足部垂直速度以及足部离地间隙。该特权向量（每帧 73 维）堆叠 5 帧后，与地形高度测量值（187 个点）拼接，总共得到 552 维。

2）动作空间：策略输出一个 12 维动作向量 $a_t \in \mathbb{R}^{12}$，对应于 12 个主动自由度（每条腿 6 个：髋关节俯仰、髋关节横滚、髋关节偏航、膝关节、踝关节俯仰、踝关节横滚）的目标关节位置偏移。目标关节位置计算如下：

$$q_{\text{target}} = a_t \cdot k_{\text{action}} \cdot k_{\text{gain}} + q_{\text{default}} \tag{4}$$

其中 k_{action} 是动作缩放因子（行走/爬楼梯时为 0.25，跑步时为 0.3，正步走时为 0.4），$k_{\text{gain}} = \tau_{\max} / K_P$ 是逐关节增益，用于根据关节的刚度和力矩限制对动作输出进行归一化。

目标位置被发送至 PD 控制器：

$$\tau = K_P(q_{\text{target}} - q) - K_D\dot{q} \tag{5}$$

比例增益 K_P 在行走型步态中为 75 至 280 N·m/rad（髋关节俯仰和膝关节为 280，髋关节滚转为 240，髋关节偏航为 75，踝关节为 75），微分增益 K_D 为 5 至 14 N·m·s/rad。正步走采用显著更高的刚度（髋关节俯仰 450，膝关节 800，髋关节滚转 300，髋关节偏航 150，踝关节 200）和阻尼（膝关节上的K_D 最高达 40），而跑步则适度增加（髋关节俯仰和膝关节为 300，髋关节滚转 240，髋关节偏航 100，踝关节 80）。计算所得力矩被裁剪至 URDF 的 effort 字段中指定的每关节最大限值（60–150 N·m）以内。策略以 50 Hz 执行，PD 控制器以 200 Hz 运行（抽取因子为 4）。指数移动平均滤波器对动作进行平滑处理，5–10 个时间步的随机动作延迟用于模拟真实世界的延迟。

3）奖励函数：所有五种步态共享统一的奖励公式。每个时间步的总奖励为加权和：

$$r_{\text{total}} = \sum_i w_i r_i \tag{6}$$

奖励项分为四类：

（1）关节跟踪奖励。主要目标是跟踪参考轨迹。对于周期性步态，髋关节俯仰和膝关节的跟踪奖励采用双指数和的形式计算：

$$r_{\text{track}} = 0.15 \sum_{j \in \{\text{hip,knee}\}} \sum_{i \in \{\text{L,R}\}} \left(e^{-4e_{ij}^2} + e^{-20e_{ij}^2} \right) \tag{7}$$

其中 $e_{ij} = q_{ij} - q_{\text{ref}}$、$_{ij}$ 为关节误差，膝关节项被掩蔽，仅在摆动相期间起作用。行走和跑步使用通用跟踪权重 $w_{\text{track}} = 2.0$，正步走使用 4.0（具有更紧的 $\sigma = 0.15$，而非 0.2）；另外，单独的仅膝关节奖励增加 $w = 3.0$（行走）或 25.0（正步走）。对于跳跃，单个跟踪奖励（w =20.0）计算所有 12 个关节的均方误差。

(2) 基座平衡与姿态奖励。包括朝向奖励（行走和跑步为 $w = 3$，正步走为 5.0，跳跃为 15.0）、基座高度奖励（$w = 2$），以及线速度（行走：$w = 5.0$，正步走：8.0，跑步：20.0，跳跃：0.0）和角速度（$w = 3.5$）的跟踪。

（3）动作平滑度奖励。动作平滑度对一阶和二阶变化（$w = -0.01$ 用于行走和跑步，-0.001 用于正步走）、关节速度（$w = -10^{-4}$）和额定扭矩（$w = -0.02$ 每侧用于行走，-0.03 用于正步走，-0.05 用于跑步）施加惩罚。正步走额外包含腿部伸直度（$w = 12.0$）、小腿抬起（$w = 8.0$）和踢脚（$w = 6.0$）项。

（4）安全性与步态特定奖励。当机体接触地面，或 |滚转角| > 0.8 弧度、 |俯仰角| >1.0弧度时，触发摔倒终止。对膝关节碰撞施加惩罚。周期性步态包括足部摆动高度（$w = 2$）和交替抬脚（$w = 6.0$ 用于正步走）。跳跃使用专门奖励：跳跃高度目标（$w = 150.0$，以0.275m为中心的高斯奖励，并带有站立阶段惩罚，以防止虚假的小幅度跳跃）、垂直起跳速度（$w = 120.0$，分段非线性：低于1.0 m/s时为线性上升，二次

above), and synchronized feet contact ($w = 20.0$), while penalizing horizontal velocity ($w = -15.0$).

The key design principle is that *all five gaits use the same reward formulation*—diversity arises from reference specification, weight tuning, and gait-specific terms rather than structural differences.

C. Selective AMP Strategy

1) Brief Review of AMP: Adversarial Motion Prior (AMP) combines adversarial imitation learning with reinforcement learning to produce motions that resemble a set of reference demonstrations. The method introduces a discriminator network D_ϕ trained to distinguish between state transitions from expert motion clips and state transitions produced by the policy during execution.

The discriminator architecture is a 3-layer MLP with hidden dimensions $[1024, 512, 256]$ and ReLU activations. The input to the discriminator is the concatenation of the current state s_t and the next state s_{t+1}, capturing a single state transition. The discriminator outputs a scalar logit $D(s_t, s_{t+1})$, which is converted into a reward signal:

$$r_{\text{AMP}} = \alpha \cdot \max\left(0, 1 - \frac{1}{4}(D(s_t, s_{t+1}) - 1)^2\right) \quad (8)$$

where α is a per-gait coefficient scaling the AMP reward magnitude. This reward is then linearly interpolated with the task reward:

$$r_{\text{combined}} = (1 - \beta) \cdot r_{\text{AMP}} + \beta \cdot r_{\text{task}} \quad (9)$$

where $\beta \in [0, 1]$ controls the balance between AMP regularization and task optimization.

The discriminator is trained with a binary cross-entropy loss combined with a gradient penalty for stability:

$$\mathcal{L}_D = \mathbb{E}_{\text{policy}}[(D+1)^2] + \mathbb{E}_{\text{expert}}[(D-1)^2] + \lambda \cdot \mathbb{E}_{\text{expert}}[\|\nabla_{(s,s')} D\|^2] \quad (10)$$

where the gradient penalty coefficient $\lambda = 10$. The reference motion data consists of captured human locomotion sequences, processed into 200,000 preloaded state transitions.

2) Our Selection Criterion: The central insight of our approach is that AMP's regularization effect is beneficial for some gaits but detrimental for others. We propose a per-gait selection criterion based on motion characteristics:

AMP is applied to: Walking, goose-stepping, and stair climbing. These gaits are periodic with well-defined, repeating motion cycles and moderate joint amplitudes, where stability is the primary concern. For these gaits, the reference motion distribution closely matches the desired policy output, and the adversarial reward provides a stabilizing inductive bias that accelerates convergence, suppresses erratic behavior, and produces smoother joint trajectories compatible with real actuator dynamics. The AMP coefficient α and interpolation β are tuned per gait: walking and stair climbing use $\alpha = 0.3$, $\beta = 0.8$; goose-stepping uses $\alpha = 0.6$, $\beta = 0.7$ (higher α to enforce the distinctive straight-leg style; lower β to allow more AMP influence).

AMP is NOT applied to: Running and jumping. These gaits involve large joint excursions that may exceed the range of available reference data, and require rapid acceleration and explosive dynamics. For running, setting $\alpha = 0.0$ and $\beta = 1.0$ removes AMP entirely, giving the policy full freedom to discover high-speed strategies using the actuator's complete torque capacity (up to 150 N·m on knee joints). Similarly, jumping uses $\alpha = 0.0$, $\beta = 1.0$, and disables the symmetry loss (sym_loss = False) to allow asymmetric force generation during takeoff. In both cases, the AMP discriminator would penalize the policy for deviating from the reference distribution, actively discouraging the motion extremes needed for sufficient speed or jump height.

In summary: *AMP improves stability but limits expressiveness; we select its use adaptively based on gait characteristics.* Table I summarizes the configuration for each gait.

TABLE I
SELECTIVE AMP CONFIGURATION PER GAIT.

Gait	α	β	Sym Loss
Walking	0.3	0.8	True
Goose-stepping	0.6	0.7	True
Running	0.0	1.0	True
Stair climbing	0.3	0.8	True
Jumping	0.0	1.0	False

D. Training Pipeline

1) Simulation Training: All policies are trained using the Proximal Policy Optimization (PPO) algorithm [9] in the Isaac Gym physics simulator [5] with 4096 parallel environments. The actor network uses an MLP with hidden layers [512, 256, 128] and ELU activation. The critic network shares the same architecture. The PPO algorithm runs 5 learning epochs over 4 mini-batches per iteration, with a learning rate of 5×10^{-4}, discount factor $\gamma = 0.99$, and GAE parameter $\lambda = 0.95$. The KL divergence target is set to 0.01 with an adaptive learning rate schedule. Training proceeds for up to 40,000 iterations, with the empirical observation normalizer updated online during training [10].

2) Domain Randomization: To bridge the simulation-to-reality gap, we apply extensive domain randomization during training. The following parameters are randomized at the environment level each episode:

- **Contact properties**: Static and dynamic friction uniformly sampled from $[0.2, 1.5]$; restitution from $[0.0, 1.0]$.
- **Base mass**: Perturbed by $[-2, +5]$ kg from the nominal mass.
- **Link inertia**: Scaled by $[0.5, 1.8]$ independently per link.
- **Center of mass**: Displaced by $[-0.1, 0.1]$ m in each axis.
- **Motor strength**: Randomized to $[0.7, 1.0]$ (walking/stair climbing), $[0.8, 1.2]$ (running/goose-stepping), or $[0.9, 1.1]$ (jumping).

- **PD gains**: K_P scaled by $[0.8, 1.2]$, K_D scaled by $[0.8, 1.2]$.
- **Joint properties**: Joint friction $[0.3, 1.5]$, damping $[0.3, 4.0]$, armature $[0.8, 1.2]$.
- **Action lag**: Actions delayed by a random number of timesteps from $[5, 10]$, simulating control latency.
- **External disturbances**: Random impulses applied to the base with maximum linear velocity 2.0 m/s and angular velocity 1.5 rad/s, at intervals of 10 seconds.
- **Initial state**: Joint positions scaled by $[0.5, 1.5]$ and offset by $[-0.1, 0.1]$ rad at reset.

A curriculum is employed for action lag: the minimum delay is gradually increased from 2 to 5 timesteps over the course of training, starting from a milder setting to facilitate initial learning and progressively increasing the difficulty.

3) Sim-to-Real Transfer: Trained policies are deployed on the physical humanoid robot using zero-shot transfer—no fine-tuning on real hardware is performed. The deployment pipeline consists of:

1) **Policy export**: The trained actor network is exported to ONNX format with the empirical observation normalizer as a single inference module, and loaded via ONNX Runtime on the robot's onboard computer.
2) **Onboard execution**: The policy runs onboard the robot's computer at 50 Hz, receiving proprioceptive observations from an external IMU (angular velocity and orientation) and joint encoders (position and velocity). Since base linear velocity is not directly measurable on the hardware, the corresponding observation channels are set to zero, relying on the observation history stack to implicitly capture velocity information.
3) **PD control**: Joint torques are computed by the onboard PD controller at 500 Hz using the target positions output by the policy. A first-order low-pass filter with cutoff frequency 50 Hz smooths the policy actions before conversion to joint targets.

The domain randomization applied during training ensures that the policy is robust to the discrepancies between simulated and real dynamics, including differences in motor response, contact behavior, and sensor noise.

III. Experiments

The video of the experiments can be found at `https://linqi-ye.github.io/video/lite.mp4`

A. Experimental Setup

Experiments were conducted on the Lite_11 humanoid lower-body platform, which provides 12 actuated joints spanning the bilateral hip, knee, and ankle degrees of freedom. All policies were trained in Isaac Gym and then deployed on the physical robot for validation. We considered five locomotion modes: walking, goose-stepping, stair climbing, running, and jumping. Stair climbing includes both ascent and descent, whereas jumping consists of in-place two-foot takeoff followed by landing recovery. All tasks share the same policy interface: a 12-dimensional action vector, a 50-dimensional per-frame observation, and a 21-frame history stack, resulting in a 1050-dimensional policy input. Consequently, gait diversity is induced by reference trajectories, reward allocation, and motion-prior configuration rather than by changing the control architecture.

All tasks use a unified simulation and control backbone. The physics step is 0.005 s, and the policy is updated every four simulation steps, yielding an effective control period of 0.02 s. Low-level execution is handled by a PD position controller, which converts the policy-produced joint target offsets into actuation commands. PPO is used for all experiments. Except for jumping, which uses a slightly higher entropy coefficient to accommodate stronger exploration demands, the same base hyperparameters are shared across tasks. The main training settings are summarized in Table II.

Task environments and prior settings are defined according to gait characteristics. Walking, goose-stepping, and running are trained primarily on flat or low-complexity terrain, stair climbing is trained on triangle-mesh stairs with curriculum learning, and jumping is trained on flat terrain to emphasize vertical impulse generation, aerial stabilization, and landing recovery. Selective AMP is central to the experimental design: walking, goose-stepping, and stair climbing belong to the AMP group, whereas running and jumping are trained without AMP. This partition reflects the observation that periodic, stability-oriented motions benefit from motion priors, whereas highly dynamic motions depend more strongly on expressive freedom and task-driven optimization. To improve sim-to-real transfer robustness, domain randomization is applied throughout training, including friction, added mass, center-of-mass perturbation, motor strength, PD gains, inertia, joint friction, joint damping, initial joint scaling and bias, and action delay. Stair climbing uses a stronger terrain curriculum, while jumping adopts a more conservative randomization range.

B. Evaluation Metrics

To evaluate multi-gait policies from both optimization and execution perspectives, we consider joint tracking quality, convergence speed, task reliability, training stability, and posture stability. Performance is not judged solely by whether a motion emerges, but by whether the learned motion adheres to the intended structure, converges stably during training, and remains controllable during execution.

The joint tracking metric characterizes how well the policy preserves the intended kinematic organization of each gait. For walking, goose-stepping, and stair climbing, it primarily reflects coordination between stance and swing phases; for running and jumping, it reflects the ability to maintain the target motion pattern under high-speed or explosive dynamics. Convergence steps quantify the optimization cost required to reach a stable regime. Tasks that stabilize within fewer iterations are easier to optimize under the unified formulation, whereas tasks that require more iterations place stronger demands on contact switching, posture recovery, or temporal coordination.

At execution time, success rate and fall rate describe task reliability. A high success rate indicates that the robot can

TABLE II

MAIN TRAINING CONFIGURATIONS FOR THE FIVE GAIT TASKS.

Gait	# Envs	Obs. Dim.	Act. Dim.	Sim Step	Control Period	num_steps_per_env	Max Iterations	AMP Group
Walking	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Goose-stepping	4096	1050	12	0.005 s	0.02 s	24	20001	AMP
Stair climbing	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Running	2048	1050	12	0.005 s	0.02 s	24	40001	No AMP
Jumping	2048	1050	12	0.005 s	0.02 s	24	11000	No AMP

repeatedly sustain the desired gait, whereas a high fall rate typically indicates sensitivity to initial conditions, rhythm control, or dynamic recovery. Training error characterizes the smoothness of optimization: lower error generally corresponds to more stable parameter updates, whereas persistently large error suggests stronger oscillation and higher uncertainty. Posture stability measures whole-body control quality. Higher posture stability corresponds to smaller torso oscillation, smoother roll-pitch regulation, and more consistent phase transitions. These metrics are used below to compare both overall gait performance and the distinction between AMP and non-AMP training.

C. Results and Analysis

1) Overall Performance of Five Gaits: Under a unified policy structure, shared observation design, and common reward organization, the proposed framework learns and deploys five locomotion modes: walking, goose-stepping, stair climbing, running, and jumping. These tasks span the spectrum from low-dynamic periodic locomotion to highly dynamic explosive motion, thereby testing the scalability of the unified RL formulation across substantially different gait classes.

Qualitatively, walking yields a regular forward stepping pattern; goose-stepping exhibits the intended high leg lift and straight-knee swing; stair climbing maintains sufficient foot clearance and body posture during both ascent and descent; running produces shorter stance durations and stronger forward propulsion; and jumping realizes the full motion chain from squat preparation and rapid takeoff to aerial stabilization and landing recovery. Figure 2 presents representative real-robot image sequences for the five gaits. These results indicate that the unified framework is able not only to maintain balance, but also to generate motion patterns with clearly differentiated dynamic characteristics and stylistic features.

Table III summarizes the quantitative performance of the five gaits. The reported indicators correspond to joint tracking quality, convergence speed, task reliability, training stability, and posture stability. Because the reward composition and task difficulty differ across gaits, these values are intended to characterize training and execution behavior under the unified framework rather than to serve as strictly normalized physical quantities across tasks.

2) Effect of AMP on Different Gaits: This subsection examines the operating boundary of AMP across gait classes. In general, AMP is most effective for gaits with clear periodic structure and stability-first objectives, whereas it can become restrictive for highly dynamic and strongly explosive motions. Walking, goose-stepping, and stair climbing fall into the former category: they require stable stance–swing transitions, consistent joint coordination, and low postural fluctuation. For such tasks, motion priors reduce ineffective exploration and typically yield lower tracking error, smoother reward growth, and faster convergence.

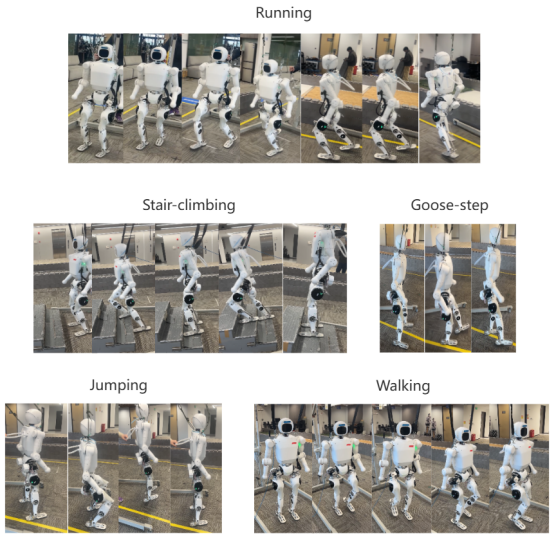


Fig. 2. Representative real-robot image sequences for the five learned gaits: walking, stair climbing, goose-stepping, jumping, and running.

Figure 3 compares goose-stepping with AMP and jumping without AMP as representative cases. Under AMP, goose-stepping exhibits a rapid increase in total reward, low training error, and a prompt decay in failure rate. The main contribution of AMP in this setting is not merely a higher terminal return, but the reinforcement of periodic structure and style consistency, which is especially important for a gait with explicit kinematic features such as high leg lift and straight-knee swing. For running, a strong AMP prior would not necessarily render the policy incapable of locomotion, but it would likely increase behavioral conservatism and bias the solution toward a more strongly regularized motion pattern. In that regime, the learned behavior would be more likely to resemble a smooth fast walk or a weak running pattern, rather than a genuinely explosive gait with clear aerial phases, strong forward propulsion, and large leg excursion. This observation suggests that, for highly dynamic locomotion, an excessively strong motion prior may improve motion regularity while simultaneously compressing the policy's access to high-amplitude dynamic solutions. By contrast, jumping

TABLE III

QUANTITATIVE SUMMARY OF THE FIVE GAIT TASKS.

Gait	Joint-Tracking Metric	Convergence Steps	Estimated Success Rate	Training Error	Posture-Stability Metric	Motion Characteristic
Walking	0.40	2500	95%	10.0	2.3	Periodic and stable forward locomotion
Goose-stepping	0.94	2500	97%	5.4	1.2	High leg lift and straight-knee swing
Stair climbing	1.00	5200	85%	1.0	0.95	Stair ascent/descent with recovery
Running	0.50	3000	96%	4.8	1.3	Short stance and strong propulsion
Jumping	15.5	3000	83%	6.0	8.6	Two-foot takeoff and landing recovery

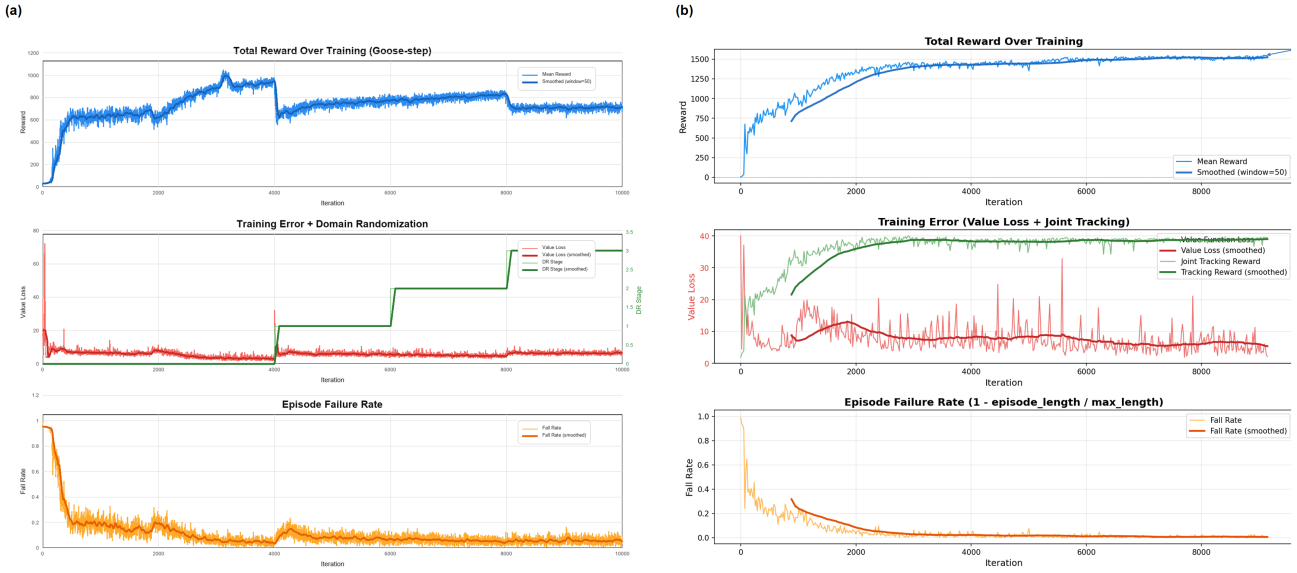


Fig. 3. Training curves used in the AMP-versus-no-AMP comparison. (a) Goose-stepping with AMP. (b) Jumping without AMP. Each panel reports total reward, training error or auxiliary metric, and failure rate.

depends on large motion amplitude, vertical impulse, and aerial freedom. Under the non-AMP setting, the task still attains sustained reward growth and low failure rate while the joint-tracking signal improves steadily. This indicates that task rewards alone are sufficient to drive effective takeoff, flight, and landing recovery when expressive exploration is preserved. These results support the view that AMP should be treated as a gait-dependent mechanism rather than a uniformly active module.

3) Real-Robot Robustness and Practical Performance: Beyond single demonstrations, the engineering value of the method is reflected in sustained execution on the physical robot. We therefore assess real-robot robustness from three perspectives: long-horizon continuity, consistency across repeated trials, and recovery under mild external disturbances or start–stop transitions.

The robustness criteria vary across gaits. For walking and goose-stepping, the emphasis lies in rhythm consistency and controlled torso oscillation; for stair climbing, in foot clearance, step placement, and posture recovery after stair transitions; for running, in stability during rapid short-stance switching; and for jumping, in synchronized takeoff, aerial posture control, landing absorption, and return to the next motion cycle. In particular, the jumping reference is explicitly divided into squat, takeoff, flight, landing, and standing-recovery phases, so robustness depends not only on leaving the ground, but also on the quality of aerial motion and landing recovery.

Taken together, the real-robot deployments show that the learned policies are not limited to visually plausible motions in simulation, but exhibit the continuity and repeatability required for physical execution. This supports the practical viability of the proposed unified framework for multi-gait humanoid locomotion.

IV. CONCLUSION

This paper presented a unified reinforcement learning framework for multi-gait humanoid locomotion together with a selective AMP strategy. Within a common policy architecture and reward formulation, the proposed method learned five representative gaits, namely walking, goose-stepping, stair climbing, running, and jumping, and transferred them to a physical humanoid platform. The results indicate that motion priors are beneficial for periodic, stability-oriented gaits, but should be weakened or removed for highly dynamic behaviors that depend on expressive motion generation. More importantly, the proposed framework provides a practical and reproducible baseline for multi-gait learning on real humanoid hardware by combining a unified RL formulation, gait-dependent prior selection, and real-robot validation. Overall, this study suggests that practical humanoid mobility will depend not only on stronger policies, but also on principled methods for organizing diverse behaviors within a shared learning framework.

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